Recommendation System

Collaborative Filtering with SVD Matrix Factorization

Felipe Taha Sant'Ana

Agenda

1. Overview
2. Data Analysis
3. SVD Model
4. Recommendations
5. Conclusions

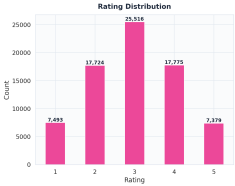
The Recommendation Problem

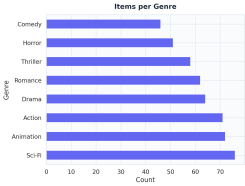
Dataset

- **1,500 users** $\times$ **500 items** (movies)
- $\sim$**75K ratings** (1–5 stars)
- **8 genres**: Action, Comedy, Drama, Sci-Fi, etc.
- Sparsity: **89.8%** of matrix is empty

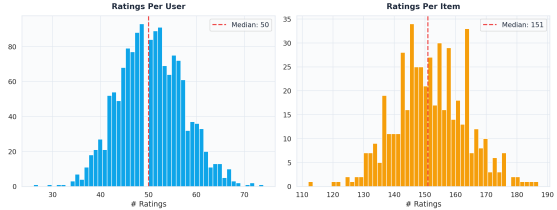
Approach

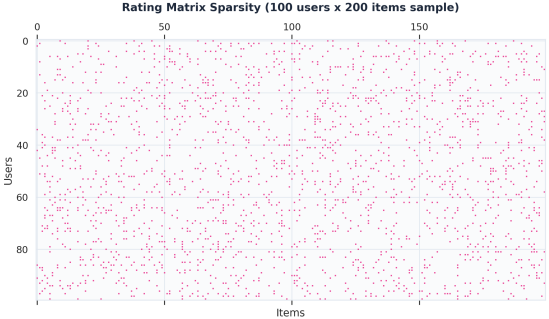
Compare baseline methods (global/user/item mean) against **SVD matrix factorization** for personalized recommendations.





Activity & Sparsity

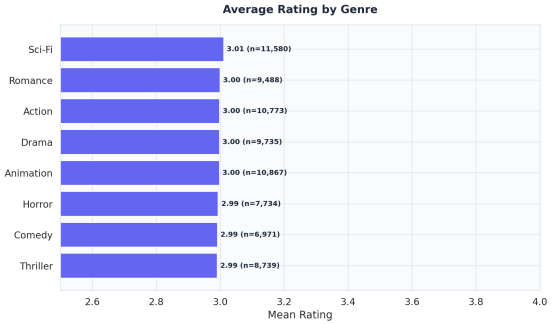


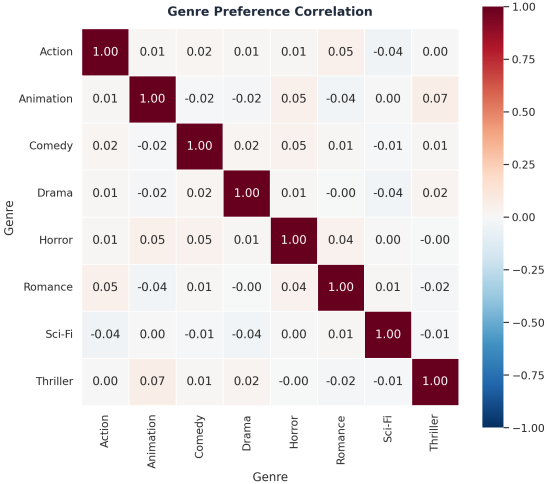


The Sparsity Challenge

Most users rate only a small fraction of items. This extreme sparsity makes collaborative filtering challenging and motivates latent factor models like SVD.

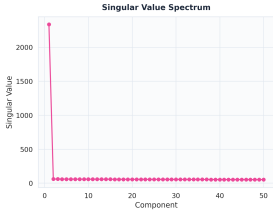
Genre Preferences & Correlations

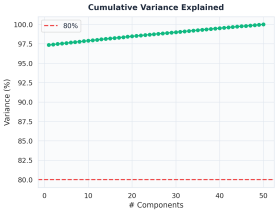


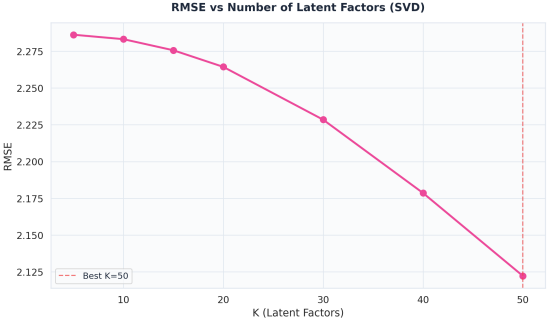


Taste Clusters

Matrix Factorization & RMSE vs K



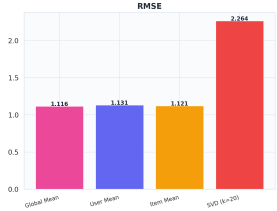


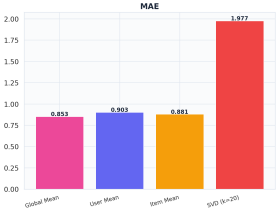


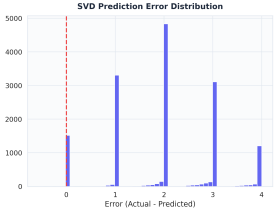
Latent Factor Analysis

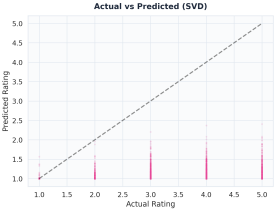
Singular value spectrum shows the first 20–30 factors capture the majority of rating structure. RMSE improves with more factors, with diminishing returns beyond $K \approx 30$.

Model Comparison & Error Analysis

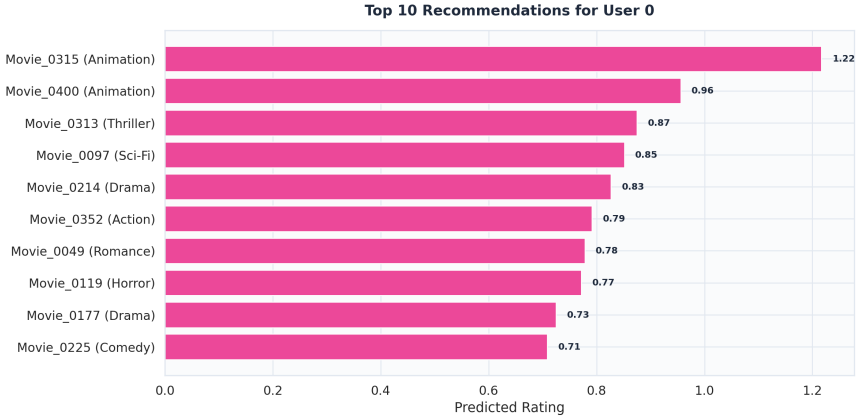








Personalized Top-N Recommendations



Output

The system generates personalized ranked lists of unseen items for each user, with predicted ratings reflecting learned latent preferences.

Summary & Future Work

Achievements

- SVD outperforms mean baselines
- K=20–30 factors optimal
- Genre correlations confirm taste structure
- Personalized top-N generation

Future Work

1. ALS for implicit feedback
2. Neural Collaborative Filtering
3. Hybrid content + CF models
4. Cold-start strategies
5. Online A/B testing of recommendations

Thank You

Questions & Discussion

GitHub: github.com/ftahas
Email: ftahas@proton.me
LinkedIn: linkedin.com/in/ftsantana